

Natural Computing SoSe25

Exercise Sheet 3 (Solution) — June 5th, 2025

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1 Quantum Computing Basics

(a) Which of the following are valid quantum states? For each valid state among the formulas below, give the probabilities of observing 0 and 1 when the system is measured in the standard computational basis.

(i) $|a\rangle = \frac{1}{\sqrt{2}}(|1\rangle + |0\rangle)$

$$= \frac{1}{\sqrt{2}}|1\rangle + \frac{1}{\sqrt{2}}|0\rangle$$

$$\langle 0|a\rangle = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$$

$$\langle 1|a\rangle = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$$

$$\rightarrow \frac{1}{2} + \frac{1}{2} = 1 \text{ (valid state)}$$

$$\Pr(M(|a\rangle) = 0) = 0.5, \Pr(M(|a\rangle) = 1) = 0.5$$

(ii) $|b\rangle = 0.6|0\rangle + 0.4|1\rangle$

$$\langle 0|b\rangle = |0.6|^2 = 0.36$$

$$\langle 1|b\rangle = |0.4|^2 = 0.16$$

$$\rightarrow 0.36 + 0.16 = 0.52 \neq 1 \text{ (invalid state)}$$

(iii) $|c\rangle = \frac{\sqrt{3}}{2}i|0\rangle - \frac{1}{2}|1\rangle$

$$\langle 0|c\rangle = \left|\frac{\sqrt{3}}{2}i\right|^2 = \left|\frac{3}{4}(-1)\right| = \frac{3}{4}$$

$$\langle 1|c\rangle = \left|-\frac{1}{2}\right|^2 = \frac{1}{4}$$

$$\rightarrow \frac{3}{4} + \frac{1}{4} = 1 \text{ (valid state)}$$

$$\Pr(M(|c\rangle) = 0) = \frac{3}{4}, \Pr(M(|c\rangle) = 1) = \frac{1}{4}$$

* that the absolute value for complex numbers $z = a + bi$ is their magnitude, and, the following holds: $|z|^2 = |z^2|$.

(Taken from www.cl.cam.ac.uk/teaching/exams/pastpapers/y2010p9q11.pdf)

(b) Write the state $|\psi\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{i}{\sqrt{2}}|01\rangle$ as two separate qubits.

The general 2-qubit-product-state for $|\psi_0\rangle = a|0\rangle + b|1\rangle$ and $|\psi_1\rangle = c|0\rangle + d|1\rangle$ is given via $|\psi_0\psi_1\rangle = |\psi_0\rangle \otimes |\psi_1\rangle$

$$= (a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle) = \begin{bmatrix} a \times \begin{bmatrix} c \\ d \end{bmatrix} \\ b \times \begin{bmatrix} c \\ d \end{bmatrix} \end{bmatrix} = ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle.$$

In order to be correct the parameters should look like this:

$$\frac{1}{\sqrt{2}}|00\rangle + \frac{i}{\sqrt{2}}|01\rangle + 0|10\rangle + 0|11\rangle$$

Since we know that neither of the factors a, c , or d can be zero (since the states $|00\rangle$ and $|01\rangle$ are non-zero) and the states $|10\rangle$ and $|11\rangle$ are zero, the factor b has to be 0. This implies $|a|^2 = 1$. In other words, the initial qubits might look like this:

$$|\psi_0\rangle = 1|0\rangle + 0|1\rangle$$

$$|\psi_1\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle$$

(Taken from the qiskit textbook chapter on multiple qubits and entangled-states.^a)

^a<https://learn.qiskit.org/course/ch-gates/multiple-qubits-and-entangled-states#ex1>

2 Simple Quantum Circuits

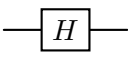
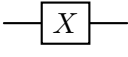
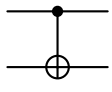
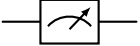
Operator	Gate	Matrix
Identity (I)		$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
Hadamard (H)		$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
Pauli-X (X)		$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Controlled-X (CNOT)		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$
Measure		

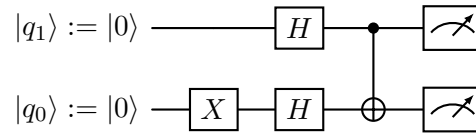
Table 1: Quantum Gate Operators

In the lecture we have introduced our abridged but complete definition of quantum com-

puting (QC). Let's try to apply our formal understanding to some practical examples.¹

(a) Using the quantum operators we have covered in the lecture (cf. Table 1), *calculate* the expected measurement probabilities.

Hint: We can compute the application of a 2-qubit gate G to the qubits q_1, q_0 via $G|q_1q_0\rangle$.



¹But you may of course still consult <https://quantum.ibm.com/composer>. However, note that the IBM Quantum Composer sorts the qubit wires differently.

We'll work through the operations step by step.

1. Let initial State for $q_1 = \phi$ and $q_0 = \psi$ is: $|\phi_0\rangle = |\psi_0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

2. Apply X gate on ψ_0 :

$$X|\psi_0\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} |\psi_0\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle = |\psi_1\rangle$$

3. Apply H gate on ϕ_1, ψ_1 separately (which we can do since ϕ_1, ψ_1 are still unentangled):

$$H|\phi_1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = |\phi_2\rangle$$

$$H|\psi_1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix} = |\psi_2\rangle$$

4. Apply $CNOT$ on $\phi_2 \otimes \psi_2$:

$$|\phi_2 \otimes \psi_2\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \otimes \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} 0.5 \\ -0.5 \\ 0.5 \\ -0.5 \end{pmatrix}$$

$$CX|\phi_2 \otimes \psi_2\rangle = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} |\phi_2 \otimes \psi_2\rangle = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0.5 \\ -0.5 \\ 0.5 \\ -0.5 \end{pmatrix} = \begin{pmatrix} 0.5 \\ -0.5 \\ -0.5 \\ 0.5 \end{pmatrix}$$

In other words, we are left with: $|\phi_3 \otimes \psi_3\rangle = 0.5|00\rangle - 0.5|01\rangle - 0.5|10\rangle + 0.5|11\rangle$.

The measurement probabilities are thus:

$$\Pr(M(|\phi_3 \otimes \psi_3\rangle) = |00\rangle) = |0.5|^2 = 0.25$$

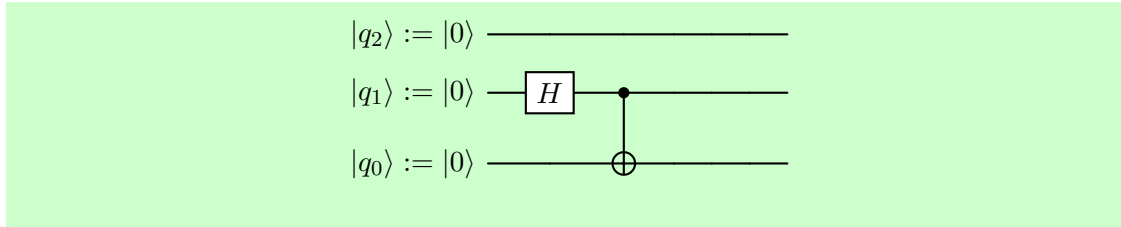
$$\Pr(M(|\phi_3 \otimes \psi_3\rangle) = |01\rangle) = |-0.5|^2 = 0.25$$

$$\Pr(M(|\phi_3 \otimes \psi_3\rangle) = |10\rangle) = |-0.5|^2 = 0.25$$

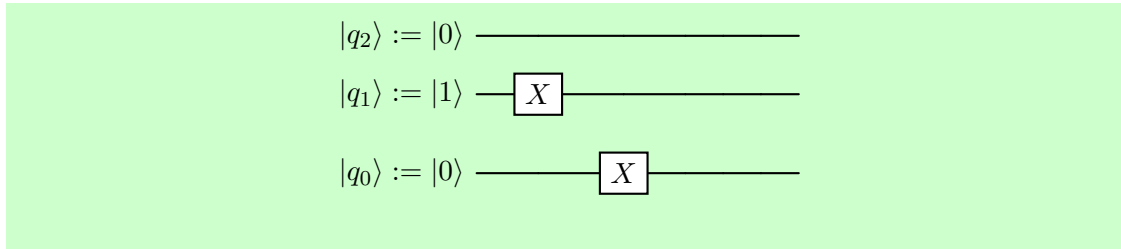
$$\Pr(M(|\phi_3 \otimes \psi_3\rangle) = |11\rangle) = |0.5|^2 = 0.25$$

(b) Complete the circuits with the gates given in Table 1 so that the following states would be prepared (given that we measure all states):

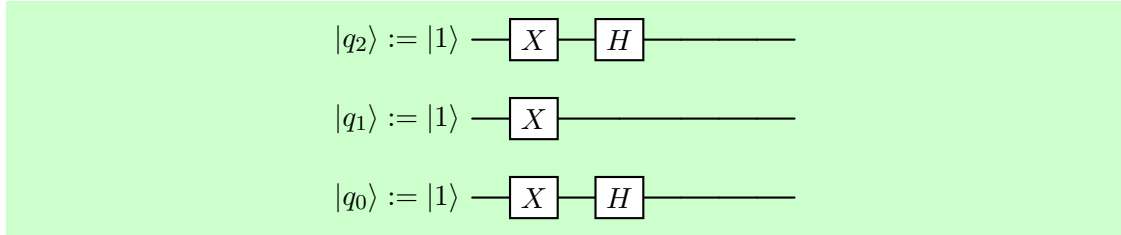
(i) $\frac{1}{\sqrt{2}}|000\rangle + \frac{1}{\sqrt{2}}|011\rangle$



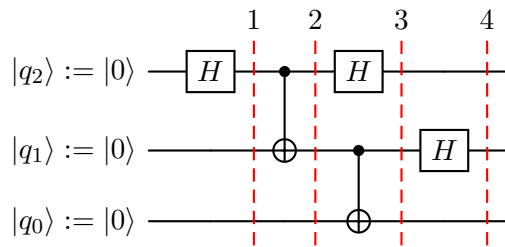
(ii) $|001\rangle$



(iii) $\frac{1}{\sqrt{4}} |000\rangle + \frac{1}{\sqrt{4}} |001\rangle + \frac{1}{\sqrt{4}} |100\rangle + \frac{1}{\sqrt{4}} |101\rangle$



(c) (Bonus) What final state distribution can we expect the following circuit to produce? What are the respective quantum states (i.e., amplitudes) at the end of the four sections?



Let's denote the states after each section of the circuit with $\psi_1, \dots, \psi_4$. With the given circuit and wire order $|q_2q_1q_0\rangle$, we would measure:

1. Equal superposition on q_2 :

$$\psi_1 \rightarrow \frac{1}{\sqrt{2}} |000\rangle + \frac{1}{\sqrt{2}} |001\rangle,$$

2. an entanglement of the first two qubits after computing $(CNOT \otimes I) |\psi_1\rangle$ i.e.,

$$\psi_2 \rightarrow \frac{1}{\sqrt{2}} |000\rangle + \frac{1}{\sqrt{2}} |011\rangle,$$

3. further entanglement of the last two qubits, needs an 8×8 operator now. We calculate $(H \otimes CNOT) |\psi_2\rangle$ for

$$\psi_3 \rightarrow \frac{1}{\sqrt{4}} |000\rangle + \frac{1}{\sqrt{4}} |001\rangle + \frac{1}{\sqrt{4}} |110\rangle - \frac{1}{\sqrt{4}} |111\rangle,$$

4. and finally, add another Hadamard gate on the whole state with $(I \otimes H \otimes I) |\psi_3\rangle$, resulting in an equal superposition over all three qubits:

$$\psi_4 \rightarrow \frac{1}{\sqrt{8}} (|000\rangle + |001\rangle + |010\rangle + |011\rangle + |100\rangle - |101\rangle - |110\rangle + |111\rangle)$$